

PRD — MyGameON Hub: Game Detail Page & Dual-Link System

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1. Executive Summary

Konteks

MyGameON Hub adalah platform game discovery + installation service yang saat ini belum punya halaman detail per-game. User landing di katalog, klik card, tapi belum ada destination yang proper untuk konversi. Hal ini membatasi:

- Kemampuan menjual layanan instalasi (Shopee package) secara optimal
- Positioning sebagai "trusted gaming hub" — bukan distributor abu-abu
- Pengalaman user yang ingin info lengkap sebelum keputusan beli

Apa yang Dibangun

Game Detail Page — halaman lengkap per game dengan:

- Pattern 2 dual-link (Shopee primary + official platforms secondary)
- Content blueprint lengkap (cover, screenshots, description, system requirements, dst)
- Honest disclosure soal capability file vs versi resmi
- Auto-generated content untuk efisiensi solo dev
- Version tracking dengan Steam Web API

Mengapa Ini Penting

Selama ini website MyGameON adalah **etalase tanpa ruang sales**. User tertarik tapi nggak ada tempat untuk *commit*. Detail page mengubah ini — sekaligus jadi **proof poin etika**: dengan menampilkan link resmi developer, MyGameON memposisikan diri sebagai *service provider*, bukan distributor ilegal.

Outcome yang Diharapkan

- Konversi: dari ~visitor ke purchase di Shopee meningkat (target awal: tracking baseline)
- Trust: user paham value proposition (curation + IT support + convenience)
- Operasional: solo dev bisa add game baru dalam ~12 menit (vs sebelumnya manual semua)
- Etika: positioning yang sustainable secara legal & moral

2. Vision & Positioning

Reframing Bisnis (Kritis untuk Dipahami)

MyGameON tidak menjual game. Game adalah medium. Yang sebenarnya dijual:

Komponen	Value yang Dibayar User
Curation	File yang sudah diverifikasi bersih dari virus/malware
IT Support	Bantuan instalasi, troubleshooting, setup
Knowledge	Info versi, kompatibilitas, requirements
Convenience	Siap pakai, offline-ready, no hassle

Framing ini menentukan SEMUA decisions di PRD ini. MyGameON adalah **Gaming Concierge Service**, bukan toko game.

Dual-Link Philosophy

Pattern 2 (Shopee primary + official secondary) bukan sekadar UI choice — ini **positioning statement**:

Official Link → BELI GAME-nya (lisensi, IP, support developer)

Shopee Link → BELI JASA KAMI (curation + install + support)

Ini bukan dua produk yang bersaing. Ini dua hal berbeda yang user bisa kombinasikan. Beli legal di Steam → tetap pakai jasa MyGameON untuk install offline. Atau cuma butuh layanan MyGameON saja. **Pilihan ada di user.**

Etika & Trust

- **Tidak menjual game developer Indonesia** sebagai bentuk apresiasi industri lokal
- **Verifikasi keamanan file** sebelum disediakan
- **Atribusi cover art** dengan template branding sendiri
- **Honest disclosure** capability file (offline-only) vs game (mungkin online)

Tone & Bahasa

- Bahasa Indonesia
- Tidak kaku, tidak terlalu santai
- Hindari istilah teknis berlebihan
- Konsisten gunakan "Anda" atau "kamu"? **Saran: pakai "kamu"** — sesuai vibe gamer dan website yang sudah pakai "kamu" di copy existing

3. User Personas & Use Cases

Persona Utama: "Pengguna Game Crack yang Cari Aman"

- Demografi: laki-laki/perempuan 18-40 tahun, gamer kasual hingga hardcore
- Pain point: pernah kena virus dari situs torrent abal-abal, takut malware
- Behavior: udah paham game crack itu gimana, mereka cari **provider yang trustable**
- Budget: cukup untuk bayar Rp 10rb-30rb per game install service
- Tech literacy: bervariasi — sebagian paham instalasi, sebagian butuh tutorial
- Channel awareness: discover MyGameON via Shopee atau referral

Persona Sekunder: "Casual Gamer Cari Convenience"

- Demografi: 25-45 tahun, kerja kantoran, main game weekend
- Pain point: nggak mau ribet research game, install puluhan GB, dst
- Behavior: prefer "siap main" — bayar lebih untuk skip the hassle
- Budget: less price-sensitive
- Tech literacy: medium — bisa install tapi prefer asisten
- Channel awareness: mostly via Shopee search atau iklan

Use Case 1: Discovery → Purchase

1. User cari game di Shopee atau Google
2. Click ke MyGameON detail page (via slug URL)
3. Lihat cover, deskripsi, screenshots → tertarik
4. Lihat info paket: versi, ukuran, partsCount, packageType
5. Lihat playModes: "Singleplayer + Splitscreen" → cocok dengan kebutuhan
6. Click CTA Shopee → checkout
7. Setelah beli: install, kontak via WhatsApp jika butuh bantuan

Use Case 2: Discovery → Lebih Memilih Versi Resmi

1. User landing di detail page
2. Lihat: "Game ini support multiplayer online di versi resmi"
3. User butuh fitur online → click link Steam
4. Beli di Steam (support developer)
5. MyGameON tidak dapat sale, tapi dapat trust untuk visit berikutnya

Use case 2 ini **tetap berharga** karena membangun reputasi MyGameON sebagai sumber info honest.

Use Case 3: Discovery → Request

1. User cari game tertentu, tidak ada di katalog

2. Submit request via form existing
 3. Admin review → upload jika file tersedia
 4. Game muncul di katalog dengan detail page lengkap
 5. User notif (atau cek manual) → purchase
-

4. Database Schema

4.1 Collection: `games/{id}` — PUBLIC DATA

interface Game {

// === IDENTITY ===

id: string; // auto-generated Firestore ID

title: string; // "Crysis 2 Remastered"

slug: string; // "crysis-2-remastered" (URL-safe, unique)

// === CLASSIFICATION ===

genres: string[]; // ["action", "adventure", "shooter"]

tags: string[]; // ["atmospheric", "story-driven"]

platform: string; // "PC" — sistem operasi

// === FILE INFO ===

fileVersion: string; // "v1.3.4" — versi file di Gdrive kamu

fileEdition?: string; // "Standard" | "Deluxe" | "GOTY" | etc — optional

fileSizeBytes: number; // 51539607552 (= 48 GB)

partsCount: number; // 13

packageType: string; // "PRE-INSTALLED" | "INSTALLER-GOG" | etc

// === GAMEPLAY (FILE-MU) ===

playModes: string[]; // ["singleplayer", "splitscreen"] — apa yg file-mu support

// === MEDIA ===

coverImageUrl: string; // wajib ada sebelum publish

screenshots: string[]; // 3-5 URL gambar

videoUrl?: string; // YouTube embed URL — optional

// === DESCRIPTIVE ===

description: string; // long-form, multi-paragraf

shortDescription: string; // 1 kalimat untuk card preview

// === DUAL-LINK SYSTEM ===

shopee: {

 isAvailable: boolean;

 url: string;

 packagePrice?: number; // optional, harga paket install di Shopee

```
};
```

```
officialPlatforms: Array<{
```

```
    platform: string;    // "steam" | "gog" | "epic" | "itch" | etc
```

```
    url: string;
```

```
    isAvailable: boolean;
```

```
}>;
```

```
// === STEAM INTEGRATION ===
```

```
steamAppId?: string;    // "268910" — untuk version tracking, optional
```

```
// === RELATED GAMES (HYBRID AUTO/MANUAL) ===
```

```
relatedGameIds: string[];    // max 4 game IDs
```

```
relatedGamesMode: 'auto' | 'manual'; // default 'auto'
```

```
// === STATUS & QC ===
```

```
availabilityStatus: 'available' | 'maintenance' | 'unavailable';
```

```
isProblematic: boolean;    // sembunyikan dari publik kalau true
```

```
// === VERSION TRACKING ===
```

```
lastFileUpdatedAt: Timestamp;    // kapan file di-refresh terakhir
```

```
lastVersionCheckAt?: Timestamp;    // kapan terakhir cek Steam API
```

```
steamLastUpdatedAt?: Timestamp; // dari Steam API result
```

```
versionStatus: 'current' | 'possibly_outdated' | 'unchecked';
```

```
// === TIMESTAMPS ===
```

```
createdAt: Timestamp;
```

```
updatedAt: Timestamp;
```

```
}
```

4.2 Collection: `gamesPrivate/{id}` — ADMIN-ONLY

Same `id` sebagai `games/{id}` untuk pairing yang konsisten.

```
interface GamePrivate {
```

```
  storageLocation: string;      // "mygameon@mydrivee.cloud"
```

```
  adminNotes: string;          // "BACKUP", "JANGAN DIPAKE", dll
```

```
  verificationStatus: 'verified' | 'needs_check' | 'rejected';
```

```
  lastVerifiedAt: Timestamp;
```

```
  addedBy: string;             // user UID atau "Bulk CSV Import"
```

```
  coverSourceCredit: string;    // "From MobyGames, modified with brand frame v3"
```

```
// optional: untuk future feature integrity check
```

```
fileChecksums?: {
```

```
  md5?: string;
```

```
  sha256?: string;
```



```
};  
  
}
```

4.3 Collection: `metadata/{type}/{slug}` — REFERENCE LISTS

Sub-collections: `genres`, `tags`, `playModes`, `platforms`.

```
interface MetadataEntry {  
  
    id: string;           // sama dengan slug, e.g. "action"  
  
    label: string;        // "Action" — display name  
  
    isActive: boolean;    // bisa dimatikan tanpa hapus  
  
    isDefault: boolean;   // dari hardcoded seeds  
  
    usageCount: number;   // auto-update via Cloud Function trigger  
  
    displayIcon?: string; // optional, untuk platforms  
  
    createdAt: Timestamp;  
  
    createdBy: 'system' | string; // 'system' untuk default, atau admin UID  
  
}
```

4.4 Default Seeds untuk Metadata

Genres (18 entries):

action, adventure, rpg, strategy, simulation, sports,
puzzle, casual, indie, racing, fighting, shooter,
horror, platformer, sandbox, survival, anime, card-game

Tags (14 starter entries):

santai, ringan, atmospheric, pixel-art, hand-drawn,
cocok-untuk-anak, story-driven, replayable, short-game,

long-game, retro, modern, dark-theme, colorful

Play Modes (8 entries):

singleplayer

multiplayer-local — multiplayer di device sama

multiplayer-online — multiplayer via internet

coop-local — kooperatif lokal

coop-online — kooperatif online

splitscreen — split-screen support

pvp-local — versus lokal

pvp-online — versus online

Platforms (10 entries):

steam, gog, epic, itch, ubisoft-connect, ea-app,

microsoft-store, xbox, playstation, nintendo

4.5 Justifikasi Keputusan Schema

Decision	Alasan
Pisah <code>games/</code> dan <code>gamesPrivate/</code>	Security by design — admin data nggak bisa bocor accidentally
<code>fileSizeBytes</code> (single) bukan <code>size + sizeUnit</code>	Standard industry, gampang sort/filter, format di frontend
<code>playModes</code> represent file-mu (bukan game general)	Honest disclosure, no false advertising
<code>relatedGamesMode: 'auto' 'manual'</code>	Kontrol penuh untuk admin saat default auto kurang relevan
<code>metadata/</code> collection untuk reference	Konsistensi (no typo) + flexibility (admin bisa nambah)

Decision	Alasan
steamAppId optional	Tidak semua game ada di Steam (e.g., game legacy, GOG-exclusive)
versionStatus computed field	UI-friendly, auto-update via Cloud Function

5. Detail Page Specification

5.1 URL Pattern

/game/{slug}

Examples:

/game/crysis-2-remastered

/game/cuphead

/game/it-takes-two

5.2 Wireframe (ASCII)

[Navbar: Logo MyGameON Search Request Status]			
[← Kembali ke Katalog]			
Crysis 2 Remastered			

Cover Image	Action · Adventure · Shooter
-------------	------------------------------

| | (600×800) | |

| | | | ⓘ v1.3.4 · 48 GB · 13 part | |

PRE-INSTALLED

PAKET INSTALASI MYGAMEON

| | | | Rp 15.000 | | |

				Siap main, verified clean				
--	--	--	--	---------------------------	--	--	--	--

| | | | [Pesan di Shopee →] | | |

| | | | Beli versi resmi di: | |

		[Steam]	[GOG]	[Epic]		
--	--	---------	-------	--------	--	--

--	--	--	--

— PLAY MODES —

| ✓ Singleplayer ✓ Split-screen

| ⓘ Catatan: file kami offline-ready. Untuk multiplayer |

| online, beli versi resmi di Steam/GOG. |

— DESKRIPSI —	
[Long-form description, 2-4 paragraf]	
— SCREENSHOTS —	
[Image 1] [Image 2] [Image 3] [Image 4]	
(Klik untuk lightbox)	
— VIDEO (jika ada) —	
[YouTube embed]	
— GAME LAINNYA YANG MUNGKIN KAMU SUKA —	
[Card] [Card] [Card] [Card]	
— FAQ INSTALASI (Phase 2 — optional) —	
[Footer]	

5.3 Section Breakdown

Section 1: Hero (Above the Fold)

Tujuan: Konversi maksimal. User dalam 5 detik harus paham:

1. Game apa ini
2. Apa yang ditawarkan MyGameON (paket install)
3. Pilihan resmi yang available
4. Cara pesan (CTA Shopee)

Elements:

- **Cover Image** (kiri, ~40% width pada desktop, full width pada mobile)
- **Title + Genres** (genres sebagai chips)
- **Quick Specs:** `fileVersion` · `fileSizeBytes formatted` · `partsCount` · `packageType`
- **Pattern 2 CTA Block** (highlight color, prominent):
 - Label: "📦 PAKET INSTALASI MYGAMEON"
 - Price (jika ada `shopee.packagePrice`)
 - Tagline: "Siap main, verified clean"
 - Primary button: "Pesan di Shopee →"
- **Secondary Platforms** (icon-only atau small chips):
 - Hanya tampilkan platform di mana `isAvailable: true`
 - Format: ikon + nama platform, klik → buka URL official
 - Header copy: "Beli versi resmi di:"

Section 2: Play Modes

Tujuan: Honest disclosure tentang gameplay capability.

Elements:

- Render `playModes` sebagai badges: ✓ Singleplayer, ✓ Split-screen, dst
- Lookup label dari `metadata/playModes/{slug}`
- **Static disclosure copy** (jika `steamAppId` exists DAN game biasanya ada online mode):

① *Catatan: file kami offline-ready dan tidak support multiplayer online.
Untuk fitur online, beli versi resmi di Steam/GOG.*

Logic detection: kalau title atau tags mengandung indicator (e.g., genre "shooter", playModes hanya singleplayer/local), tampilkan note ini. Atau lebih sederhana: **selalu tampilkan static note ini di bawah playModes** sebagai standar disclosure.

Section 3: Description

Tujuan: Edukasi user tentang gameplay & atmosfer.

Elements:

- Long-form **description** (2-4 paragraf)
- Format: prose, no bullet points (kecuali memang relevan)
- Auto-generated default jika admin nggak custom — pakai template:

"{title} adalah game {genres[0]}{...} dengan ukuran file {fileSizeBytes formatted}. Game ini tersedia dalam paket instalasi siap main yang sudah diverifikasi keamanannya."

- Admin bisa override via rich text editor di dashboard

Section 4: Screenshots

Tujuan: Visual proof, bantu konversi.

Elements:

- Grid responsive: 4 kolom desktop, 2 kolom mobile
- Klik → lightbox modal
- Lazy-load
- Aspect ratio konsisten (16:9 recommended)

Section 5: Video (Conditional)

Tujuan: Highest engagement, kalau ada.

Elements:

- YouTube embed (responsive iframe)
- Hanya tampilkan kalau **videoUrl** exists
- Lazy-load (jangan auto-play)

Section 6: Related Games

Tujuan: Cross-sell, retention.

Elements:

- Header: "Game lainnya yang mungkin kamu suka"
- 4 card horizontal scroll (mobile) / grid (desktop)
- Pakai `relatedGameIds` — fetch detail dari `games/` collection
- Klik → navigate ke detail page game tersebut

Section 7: FAQ Instalasi (Phase 2 — Optional)

Tujuan: Reduce repetitive WhatsApp questions.

Elements:

- Accordion sections:
 - "Apa yang termasuk dalam paket?"
 - "Berapa lama proses instalasi?"
 - "Apakah butuh internet saat main?"
 - "Bagaimana kalau ada masalah?"
- Konten static initially, bisa di-customize per game di Phase 3+

5.4 Static Disclosure Copy (Reusable)

Copy yang konsisten tampil di bagian Play Modes:

① Catatan tentang multiplayer online:

File kami offline-ready dan tidak bisa untuk multiplayer

online. Untuk fitur online/multiplayer online, beli versi

resmi di platform di bawah ini.

Pertimbangan: tampilkan **selalu** di bagian Play Modes (untuk konsistensi dan transparansi), atau **kondisional** (hanya kalau game support online di Steam).

Saran: Tampilkan **selalu** sebagai standar disclosure — bikin user paham value proposition MyGameON dari awal.

5.5 Loading & Error States

- **Loading:** Skeleton screen yang match layout (cover placeholder, title placeholder, dst)
- **Game not found** (`slug` tidak match): Halaman 404 custom dengan suggest "Coba search ulang" atau "Request game ini"
- **Game isProblematic: true:** Treat as 404 untuk public, tetap accessible di admin dashboard

- **Game `availabilityStatus`: 'unavailable'**: Tampil dengan banner "Saat ini tidak tersedia" + CTA disabled

6. Game Card Update

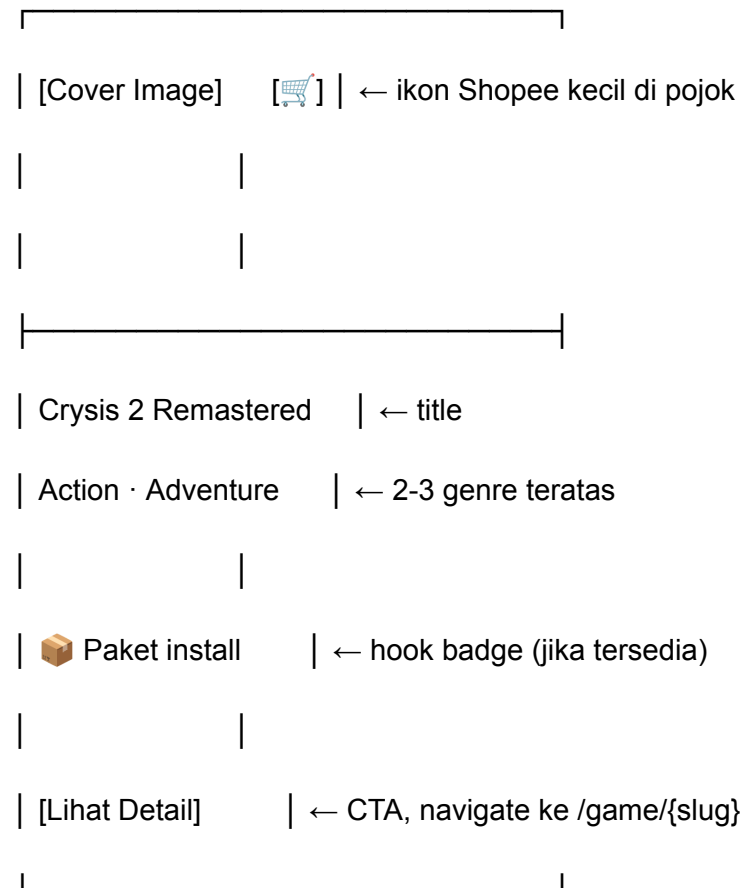
6.1 Hook Strategy

Card hanya bertugas **bikin user klik untuk masuk detail page** — bukan sebagai jalan pintas ke Shopee. Hook minimalis tapi clearly menyampaikan ada layanan.

Implementasi:

- Badge text ringkas: " Paket install tersedia" (jika `shopee.isAvailable`)
- Ikon Shopee kecil di pojok kanan atas card sebagai visual cue

6.2 Card Anatomy (Updated)



6.3 Konten Card

Element	Source Field	Logic
Cover image	<code>coverImageUrl</code>	Fallback placeholder kalau null
Title	<code>title</code>	Truncate kalau >40 chars
Genre chips	<code>genres[]</code>	Tampilkan max 2-3, lookup label dari metadata
Shopee icon	<code>shopee.isAvailable</code>	Render hanya kalau <code>true</code>
Hook badge	<code>shopee.isAvailable</code>	Text: "📦 Paket install tersedia"
CTA button	n/a	Always "Lihat Detail" → <code>/game/{slug}</code>

6.4 Apa yang TIDAK Ada di Card

- ❌ Shopee URL langsung (force user ke detail page dulu)
- ❌ Harga
- ❌ Description
- ❌ Screenshots tambahan
- ❌ Versi/file size detail (hindari clutter)

6.5 Variants

- **Default Card:** Untuk grid/list di katalog
 - **Featured Card:** Lebih besar, pakai cover image lebih prominent (untuk hero section / featured games)
 - **Compact Card:** Untuk related games di detail page (lebih kecil, image-focused)
-

7. Routing & URL Strategy

7.1 URL Patterns

Route	Purpose
/	Homepage (existing landing)
/game/{slug}	Detail page (NEW)
/game/ atau /games/	Redirect ke / atau full catalog
/search?q=...	Search results (existing Algolia)
/request-game	Request form (existing)
/request-status	Request status check (existing)
/about atau /how-it-works	Trust page (Phase 2 — kombinasi 2 section)

7.2 Slug Generation Rules

Auto-generate dari title saat create:

```
function generateSlug(title: string): string {  
  
    return title  
  
        .toLowerCase()  
  
        .replace(/['"]/g, "")    // hilangkan apostrof  
  
        .replace(/[^a-z0-9]+/g, '-') // ganti non-alphanumeric jadi dash  
  
        .replace(/^+|-+$/g, "")    // trim dash di awal/akhir  
  
        .replace(/-+/g, '-');      // collapse multiple dashes  
  
}  
  
generateSlug("Crysis 2 Remastered")  
  
// → "crisis-2-remastered"
```

```
generateSlug("DORAEMON STORY OF SEASONS Friends of the Great Kingdom")
```

```
// → "doraemon-story-of-seasons-friends-of-the-great-kingdom"
```

Uniqueness check:

- Saat create, cek apakah slug sudah ada di Firestore
- Jika ada, append number: `crysis-2-remastered-2`
- Cloud Function handle ini saat `onCreate` trigger

7.3 Slug Updates

Jika title diubah, slug **tidak otomatis berubah** — untuk preserve URL stability dan SEO. Admin bisa manual override slug via dashboard kalau memang perlu.

7.4 SEO Considerations

- `<title>` tag: `{game.title}` – MyGameON Hub
 - `<meta description>`: ambil dari `game.shortDescription`
 - Open Graph tags: cover image, title, description
 - Schema.org structured data (Phase 2): VideoGame schema dengan `name`, `genre`, `image`
-

8. Automation Strategy

8.1 Cloud Functions yang Dibutuhkan

`onGameCreate` (Trigger: `games/{id}` `onCreate`)

Tasks:

1. Generate `slug` dari `title` (dengan uniqueness check)
2. Generate `shortDescription` template jika kosong
3. Generate `relatedGameIds` (jika `relatedGamesMode`: `'auto'`)
4. Increment `usageCount` di metadata yang dipakai (`genres`, `tags`, `playModes`)
5. Set `lastFileUpdatedAt` = `createdAt`
6. Set `versionStatus`: `'unchecked'`

`onGameUpdate` (Trigger: `games/{id}` `onUpdate`)

Tasks:

1. Re-generate `relatedGameIds` jika genres berubah DAN mode `auto`
2. Update `usageCount` metadata jika genres/tags/playModes berubah
3. Sync ke Algolia index untuk search
4. Update `updatedAt` field

`onGameDelete` (Trigger: `games/{id}` onDelete)

Tasks:

1. Decrement `usageCount` metadata
2. Remove dari Algolia index
3. Delete pasangan dokumen `gamesPrivate/{id}`
4. Remove ID dari `relatedGameIds[ ]` di game lain (cleanup)

`checkSteamVersion` (HTTPS Callable)

Trigger: Manual dari admin dashboard, satu game atau batch.

Tasks:

1. Ambil `steamAppId` dari game
2. Call Steam Web API:
`https://store.steampowered.com/api/appdetails?appids={steamAppId}`
3. Extract `release_date` atau timestamp terbaru
4. Update `steamLastUpdatedAt` dan `lastVersionCheckAt`
5. Compute `versionStatus`:

if (`steamLastUpdatedAt` > `lastFileUpdatedAt` + 7 hari)

→ 'possibly_outdated'

else

→ 'current'

`weeklyVersionCheckBatch` (Scheduled, runs every Sunday)

Tasks:

1. Query semua game dengan `steamAppId` exists
2. Loop, call `checkSteamVersion` untuk masing-masing
3. Rate limit: 1 request per 2 detik (respect Steam API)
4. Log results ke admin notification

`syncMetadataUsageCount` (Scheduled, runs daily)

Tasks:

1. Re-compute `usageCount` untuk setiap metadata entry
2. Self-healing untuk drift dari triggers

`regenerateRelatedGames` (HTTPS Callable, optional)

Tasks:

1. Untuk semua game dengan `relatedGamesMode: 'auto'`
2. Re-compute `relatedGameIds` based on current genre similarity
3. Run secara on-demand kalau admin merasa related games udah stale

8.2 Auto-Generation Logic

Slug Generation

Lihat section 7.2.

Short Description Template

```
function generateShortDescription(game: Game): string {
```

```
  const genreLabel = game.genres
```

```
    .slice(0, 2)
```

```
    .map(g => metadataLookup('genres', g).label)
```

```
    .join(' & ');
```

```
  const sizeFormatted = formatFileSize(game.fileSizeBytes);
```

```
  return `${game.title} — game ${genreLabel} berukuran ${sizeFormatted}. Tersedia paket instalasi siap main.`;
```

```
}
```

Related Games Algorithm (for **auto** mode)

```
function findRelatedGames(targetGame: Game, allGames: Game[]): string[] {

  const candidates = allGames

    .filter(g => g.id !== targetGame.id)

    .filter(g => !g.isProblematic)

    .filter(g => g.availabilityStatus === 'available')

    .map(g => ({

      id: g.id,

      score: calculateGenreOverlap(targetGame.genres, g.genres)

    }))

    .filter(c => c.score > 0)

    .sort((a, b) => b.score - a.score)

    .slice(0, 4);

  return candidates.map(c => c.id);

}

function calculateGenreOverlap(a: string[], b: string[]): number {

  const intersect = a.filter(g => b.includes(g));

  return intersect.length;

}
```

8.3 File Size Format Helper

```
function formatFileSize(bytes: number): string {
```

```
if (bytes < 1024 ** 2) return `${(bytes / 1024).toFixed(1)} KB`;

if (bytes < 1024 ** 3) return `${(bytes / 1024 ** 2).toFixed(1)} MB`;

return `${(bytes / 1024 ** 3).toFixed(2)} GB`;

}
```

9. Security Rules Outline

9.1 Firestore Rules

```
rules_version = '2';
```

```
service cloud.firestore {
```

```
  match /databases/{database}/documents {
```

```
    // PUBLIC: anyone can read, only admin can write
```

```
    match /games/{gameId} {
```

```
      allow read: if resource.data.isProblematic == false
```

```
        && resource.data.availabilityStatus != 'unavailable';
```

```
    // OR: allow read kalau admin (untuk preview)
```

```
    allow read: if isAdmin();
```

```
    allow create, update, delete: if isAdmin();
```

```
  }
```



```
// PRIVATE: only admin
```

```
match /gamesPrivate/{gameId} {
```

```
  allow read, write: if isAdmin();
```

```
}
```

```
// METADATA: anyone can read, only admin can write
```

```
match /metadata/{type}/{slug} {
```

```
  allow read: if true;
```

```
  allow write: if isAdmin();
```

```
}
```

```
// REQUESTS: existing rules (don't change)
```

```
match /requests/{requestId} {
```

```
  // [keep existing rules]
```

```
}
```

```
// Helper function
```

```
function isAdmin() {
```

```
  return request.auth != null
```

```
    && request.auth.token.admin == true;
```

```
}
```

```
}
```

```
}
```

9.2 Cloud Storage Rules

- Cover images & screenshots: public read
- Admin upload only: validate file size, MIME type, max dimensions

9.3 Cloud Functions Auth

- HTTPS Callable functions (`checkSteamVersion`, `regenerateRelatedGames`): require admin claim
 - Triggered functions: server-side, no auth needed but validate context
-

10. Implementation Phases & Milestones

Phase 0: Foundation (Setup + Migration) — 1 hari

Goals: Schema siap, data sample siap.

- ☐ Setup `metadata/` collection dengan default seeds
- ☐ Migrate existing `games/` documents:
 - Rename fields (`genre` → `genres`, `numberOfParts` → `partsCount`, dll)
 - Convert `size + sizeUnit` → `fileSizeBytes`
 - Add new fields dengan default values
- ☐ Bikin `gamesPrivate/{id}` untuk dokumen existing
- ☐ Update Firestore security rules
- ☐ Test akses publik vs admin

Phase 1: Detail Page MVP — 2-3 hari

Goals: Detail page live dengan core sections.

- ☐ Setup route `/game/:slug`
- ☐ Bikin halaman `GameDetailPage.jsx`:
 - Hero section (cover, title, specs, Pattern 2 CTA, secondary platforms)
 - Play Modes section dengan static disclosure
 - Description section
 - Screenshots grid + lightbox
- ☐ Slug-based fetching (query Firestore where `slug == slug`)

- ☐ Loading & error states
- ☐ Mobile responsive
- ☐ Deploy & test

Phase 2: Card Update + Bonus Sections — 1-2 hari

Goals: Game card hook + nice-to-have sections.

- ☐ Update card component dengan hook badge + Shopee icon
- ☐ Add Video section (YouTube embed)
- ☐ Add Related Games section
- ☐ Test navigation flow: katalog → card → detail → related → detail

Phase 3: Admin Dashboard untuk Detail Page — 2-3 hari

Goals: Solo dev workflow optimal.

- ☐ Form add/edit game dengan field structure baru
- ☐ Autocomplete dropdown untuk genres/tags/playModes (with "add new" capability)
- ☐ Cover image uploader (Cloudinary integration)
- ☐ Screenshot uploader (multi-file)
- ☐ Rich text editor untuk description
- ☐ Toggle: Auto vs Manual related games
- ☐ Display existing `gamesPrivate` data dalam form

Phase 4: Cloud Functions — 1-2 hari

Goals: Automation jalan.

- ☐ `onGameCreate` trigger
- ☐ `onGameUpdate` trigger
- ☐ `onGameDelete` trigger
- ☐ `syncMetadataUsageCount` scheduled
- ☐ Test end-to-end: create game → triggers fire → data populated correctly

Phase 5: Steam Integration — 1-2 hari

Goals: Version tracking jalan.

- ☐ `checkSteamVersion` callable function
- ☐ Admin dashboard: button "Cek Sekarang" per game
- ☐ Display `versionStatus` badge di admin dashboard

- ☐ `weeklyVersionCheckBatch` scheduled function
- ☐ (Public) Subtle indicator di detail page kalau `versionStatus: 'possibly_outdated'`

Phase 6: Polish & SEO — 1 hari

Goals: Siap untuk lift `noindex`.

- ☐ Meta tags (title, description, OG)
- ☐ Schema.org structured data
- ☐ Performance audit (Lighthouse)
- ☐ Cross-browser test
- ☐ Mobile UX final pass
- ☐ Sitemap.xml
- ☐ Lift `noindex` setelah QA pass

Phase 7: About / How-It-Works Page (Branding) — 1 hari

Goals: Trust signals lengkap.

- ☐ `/about` page dengan 2 section:
 - "About MyGameON" — siapa kami, filosofi, support developer Indonesia
 - "How It Works" — proses verifikasi, dual-link strategy, value layanan
- ☐ Link dari navbar dan footer

Total Estimated Time: ~10-15 hari (solo dev, part-time)

11. Open Questions & Future Considerations

11.1 Open Questions (Belum Diputuskan)

Q1: Bagaimana handle game yang `availabilityStatus` berubah dari 'unavailable' → 'available'?

- Notifikasi user yang pernah request? (butuh subscription system)
- Email blast? (butuh email infrastructure)
- Skip dulu, manual notif via WhatsApp?

Q2: Apakah `videoUrl` perlu support platform selain YouTube?

- YouTube cukup untuk MVP

- Future: Vimeo, direct upload?

Q3: Bagaimana display kalau admin hapus genre/tag yang masih dipakai game?

- Saran: cegah delete kalau `usageCount > 0`
- Atau: replace dengan "uncategorized"
- Atau: deprecate (set `isActive: false`) tanpa hapus data

Q4: Apakah perlu `description` Bahasa Inggris?

- MVP: Bahasa Indonesia saja
- Future: i18n support kalau target market expand

Q5: Halaman katalog/list — perlu update juga?

- MVP: tetap pakai existing landing page Algolia search
- Future: dedicated `/catalog` dengan filter UI (genre, playModes, dst)

11.2 Future Features (Beyond MVP)

Tier 1: Quality of Life

- **FAQ Instalasi customizable per game**
- **System Requirements section** (min/recommended specs)
- **Customer testimonial system** (collect dari Shopee reviews)
- **"Updated" indicator** untuk game yang baru di-refresh
- **Saved games** (let user bookmark games tanpa login)

Tier 2: Engagement

- **User accounts** (untuk request tracking, history, wishlist)
- **Email subscription** (notify saat game requested available)
- **WhatsApp bot** untuk customer service automation
- **Affiliate program** untuk referral

Tier 3: Scale

- **i18n support** (English version)
- **Indonesian developer partnership program** (showcase game lokal)
- **Analytics dashboard** (top requested, top selling, conversion funnel)
- **A/B testing framework** (test pattern variations)
- **Multi-currency display** (untuk official platform prices)

Tier 4: Strategic

- **Indonesian developer marketplace** — area khusus untuk game lokal yang kamu *promote* (sesuai filosofi support developer Indonesia)
- **Partnership dengan Steam/GOG** untuk affiliate revenue
- **Premium subscription** untuk customer (faster support, exclusive content)

11.3 Risks & Mitigations

Risk	Mitigation
Steam API rate limit / changes	Cache aggressively, monitor for breaking changes, fallback ke manual check
Cover art legal concerns	Pakai sumber dengan atribusi jelas (MobyGames credit), apply branding template untuk transformation
Admin dashboard learning curve	Dokumentasi ringkas + video walkthrough (1x effort)
Slug collision saat title duplicate	Auto-append number (handled di Cloud Function)
Performance saat games >1000	Lazy load, pagination, Algolia for search (already setup)
User confusion antara "paket install" dan "game license"	Disclosure copy yang jelas di detail page + about page

11.4 Success Metrics (Track After Launch)

Primary KPI:

- Conversion rate: visitor detail page → klik Shopee CTA
- Daily active users di detail pages
- Average time on detail page

Secondary KPI:

- % game dengan complete data (cover, screenshots, video)
- % game dengan `versionStatus: 'current'`
- Time-to-add-new-game (target: <12 menit)

Health Metrics:

- Page load speed (target: <2s LCP)
 - Mobile vs desktop ratio
 - Search-to-detail conversion (lewat Algolia)
 - Card click-through rate
-

Closing Notes

PRD ini adalah **living document**. Update sebagaimana implementation progresses dan insights baru muncul.

Filosofi yang diterapkan dalam PRD ini:

1. **Etika sebagai foundation** — semua decisions reinforce positioning honest service provider
 2. **Solo dev efficiency** — automation prioritized, manual effort minimized
 3. **User-first design** — honest disclosure, no false advertising
 4. **Schema discipline** — naming convention, public/private separation, structured data
 5. **Iterative delivery** — MVP first, polish later, scale based on data
-

Approvals:

- Product Owner: Adli  (via brainstorm sessions)
- Technical Architecture: Reviewed in this PRD
- Implementation: Pending start

Next Action: Begin Phase 0 (Foundation Setup).

End of PRD